

Ivan Kuo Huang

Senior Software Engineer · Cross-Platform · Native Mobile · Full Stack Developer

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📄 PROFILE SUMMARY

Senior Full Stack Mobile Engineer with 12+ years of experience designing, developing, and deploying scalable cross-platform apps across web, iOS, and Android ecosystems. Highly proficient in React, React Native, and TypeScript, with strong expertise in modern state management solutions and performance-driven frontend architecture.

Possessed advanced mobile engineering expertise spanning React Native, Expo/EAS, and native iOS and Android development. Experienced in managing full app lifecycle, from development through App Store and Play Store deployment.

Skilled in backend development using Node.js and Python, with deep experience in building RESTful and GraphQL APIs. Demonstrates strong capability in end-to-end product delivery, including system design, API architecture, cloud integration, and CI/CD automation.

A product-focused engineer with strong problem-solving skills and a track record of delivering production-grade apps independently or within cross-functional teams. Adept at bridging mobile, frontend, and backend layers to build seamless, high-performance digital products.

📁 PROFESSIONAL EXPERIENCE

Senior Software Engineer (Mobile), *Fruit Street Health* 🔗

01/2025 – Present | New York, NY

- Developed a cross-platform diabetes prevention mobile experience, implemented with **React Native**, **React Navigation**, **Expo**, **Redux**, **TypeScript**, and modular feature-based architecture.
- Modernized the mobile release pipeline using **Expo Application Services (EAS)**, environment-based app configuration, and CI workflows to improve release consistency across iOS and Android.
- Built native bridges for device integration and platform-specific workflows, implemented with **Swift**, **Kotlin**, **Objective-C** interoperability, **Gradle/CocoaPods** configuration.
- Led migration work toward a new React Native architecture to improve native performance, implemented **JSI-compatible native modules**, **Fabric Renderer**, and **Turbo Modules**.
- Improved mobile reliability and release quality by building automated testing and delivery workflows with **Jest**, **React Native Testing Library**, **Detox**, **GitHub Actions**, **TestFlight**, and **Google Play Console**.

🧠 TECHNICAL SKILLS

Programming Languages

JavaScript (ES6+), TypeScript, HTML5, CSS3, Swift, Objective-C, Kotlin, Java, Python,

Mobile Development

React Native CLI, Expo/EAS, SwiftUI, UIKit, Xcode, Android Studio, Jetpack Compose

Frontend & UX/UI Development

React (16+), Next.js, Tailwind CSS, Material UI, Context-API, Redux/RTK, Zustand, MobX, Figma, AdobeXD, WCAG

Backend Development & API Integration

Node.js, Nest.js, Express, Django, FastAPI, RESTful APIs, GraphQL, gRPC, WebSockets

Cloud Services Integration

AWS (Cognito, S3, Lambda, API Gateway), Firebase (Firestore, FCM, Crashlytics), APNs

Database

MySQL, PostgreSQL, MongoDB, SQLite

AI Integration & Code Development

AI assistants, On-device, Voice interfaces, Image processing, Cursor, Claude

Testing & QA

Jest, Detox, Mocha, Appium, XCTest, Espresso, Postman, Swagger

CI/CD & Deployment

GitHub Actions, Bitrise, EAS Workflows, Google Play Store, App Store Deployment

- Accelerated mobile feature development by using AI-assisted coding tools such as **Cursor and Claude** to generate boilerplate, refactor complex React Native/TypeScript components.
- Developed AI-assisted coaching workflows, implemented with **LLM-powered assistant APIs** and human-in-the-loop dietitian review flows.

Senior React Native Engineer, Rebet Inc.

03/2023 – 12/2024 | Boston, MA

- Built a real-time social sports betting experience, developed and deployed cross-platform mobile app using **React Native, TypeScript, Redux Toolkit, and GraphQL subscriptions**.
- Enhanced real-time user experience and notification system by integrating **Firebase Cloud Messaging, background task handling, and deep linking**, enabling instant updates.
- Improved app performance and scalability by optimizing rendering and state updates using memoization strategies, lazy loading, code splitting, and performance profiling tools - **Flipper, React DevTools**.
- Designed high-performance mobile UI architecture with reusable component libraries; leveraged **React Native Reanimated, Gesture Handler, FlatList, and modular design patterns**.
- Improved app stability and release pipeline, introduced automated error tracking, **CI/CD workflows**, and **OTA update strategies** to accelerate release cycles and reduce production crash rates.

Full Stack Engineer (Mobile & Backend), TekRevol

09/2020 – 02/2023 | Houston, TX

- Built cross-platform mobile apps for startup and enterprise clients by developing **React Native** screens, navigation flows, and shared UI components with **React Hooks, Redux and Zustand**.
- Developed secure user auth and onboarding flows using biometric authentication using Face ID, Touch ID, Android Biometrics, React Native Keychain, and secure local storage.
- Engineered backend services for mobile-first products by building scalable APIs with **Node.js, Express.js, PostgreSQL/MongoDB, Firebase and AWS**.
- Implemented local caching, background sync, and persisted state using **AsyncStorage, SQLite/Realm, Redux Persist**, and API sync strategies to deliver offline-first features for content-heavy apps.
- Developed Web3 wallet and blockchain interaction features, integrated **Web3.js** for blockchain-enabled fintech and marketplace prototypes.

Frontend Engineer, ChowNow

06/2017 – 08/2020 | Los Angeles, CA

- Built and enhanced the restaurant online ordering experience (web & mobile), implemented using **React/React Native, Redux, and TypeScript**; improved UI performance and responsiveness.
- Led the migration of legacy native mobile ordering features into a unified cross-platform solution, converting iOS/Android modules into a scalable React Native codebase.

 CORE COMPETENCIES

Mobile & Frontend Development
Backend Development & API Integration
Cloud Services & AI Integration
Team Collaboration & Communication
Project & Time Management
Deployment & Product Support
Self Motivation & Continuous Learning
Agile Startup Environment

 SKILLS RATING

HTML & CSS	
JavaScript / TypeScript	
React / React Native	
Redux / RTK (Redux Toolkit)	
Zustand	
MobX	
Expo / EAS	
Swift / Objective-C	
Kotlin / Java	
Dart & Flutter	
Node.js / Nest.js	
Python	
Django	
FastAPI	
RESTful APIs	
GraphQL	
CI/CD	
Firebase	
AWS	
Git	
Testing	
App Stores	

- Executed comprehensive unit and integration testing using **Jest** and **React Native Testing Library**, reduced post-release bugs.
- Integrated **Firebase cloud services** including FCM, Analytics, Firestore, and Cloud Functions to support real-time order updates, push notifications, and user engagement tracking.
- Participated in **agile development lifecycle**; contributed to scalable frontend architecture decisions and long-term maintainability.

Software Developer, Robinhood

01/2015 – 05/2017 | Menlo Park, CA

- Built and enhanced core mobile trading experiences; implemented across **iOS (Objective-C, early Swift adoption)** and **Android (Java)**.
- Developed web-based user dashboards and internal operational tools using **React.js** and **Node.js** alongside mobile platforms.
- Developed low-latency real-time market data streaming features to deliver live price updates and quote refreshes using **WebSockets**.

Freelance Developer, Upwork

06/2013 – 12/2014 | Remote, US

- Developed end-to-end client web apps and custom websites; built responsive UIs and backend logic.
- Implemented full stack integrations including RESTful APIs, Stripe payment processing, and WordPress/Shopify customization.

Used Skills: HTML5, CSS3, JavaScript (ES6+), PHP, Ajax, Bootstrap, MySQL, REST, Stripe, WordPress, Shopify, AWS, Git/GitHub

PROJECTS

Fruit Street Health App, Fruit Street Remote Care Mobile App

[App Store](#)  [Play Store](#) 

- Virtual Diabetes Prevention Program - class enrollment, onboarding, weight tracking, food logging, and coaching workflows
- Connected-device health tracking for wireless scale and activity data, integrated with Fitbit and Bluetooth devices
- Secure participant-to-dietitian messaging and group support features, including threaded conversations, unread states, attachment handling, push notifications
- Remote patient monitoring dashboards inside the mobile app with trend charts for weight, activity, and program milestones

Used Skills: React Native, TypeScript, Expo EAS Build, EAS Submit, iOS, Android, Swift, Kotlin, Redux Toolkit, React Hooks, Context-API, custom hooks, React Navigation, REST APIs, Fitbit API, BLE (react-native-ble-plx), FCM, APNs, React Native Testing Library, Detox, Jest, E2E testing, TestFlight, Google Play Console, Cursor, Claude

ReBet: Social Sports & Casino, Social Sweepstakes Sportsbook & Casino

[App Store](#)  [Play Store](#) 

- Real-time sports betting experience where users can place bets, track live games, and share predictions within a social feed
- Mobile wallet experience allowing users to deposit funds, withdraw winnings, and track balances in-app. (Stripe/Apple Pay/Google Pay)

Used Skills: React Native, TypeScript, Redux/Redux Toolkit, React Query, React Navigation, Axios, React Native Reanimated, Gesture Handler, MVVM, AsyncStorage, WebSockets, GraphQL, Node.js, Express, Redis, PostgreSQL, JWT, Firebase Authentication, Analytics, FCM, Biometric authentication (Face ID/Touch ID), CI/CD, Docker

EDUCATION

Master of Science (MS), Computer Science,
The University of Texas at Arlington

2016 – 2018 | Arlington, TX

Bachelor of Science (BS), Mathematics,
Rice University

2009 – 2013 | Houston, TX

CERTIFICATES

Neural Networks and Deep Learning 

DeepLearning.AI, Issued Dec 2024

INTERESTS

Blogging | Sports | Cooking

ChowNow: Local Food Ordering, *Consumer-facing app for ordering food from local restaurants*

[App Store](#)  [Play Store](#) 

- End-to-end online food ordering experience allowing users to browse restaurants, customize menu items, and place orders in real time
- Real-time tracking experience, allowing customers to view order progress; received-preparing-ready-delivered
- In-app payment experience with multiple payment methods supported; Credit cards, Debit cards, Apple Pay, Google Pay

Used Skills: React Native CLI, TypeScript, Redux for state management, React Native Testing Library, Styled Components, Axios, Swift, Java, REST, FCM for push notification, WebSocket, Google Maps API, Stripe

Robinhood: Trade Anything, *Stock & crypto trading app*

[App Store](#)  [Play Store](#) 

- Stock order placement, portfolio tracking, and account management
- Secure user authentication, onboarding, and ID verification (KYC)
- End-to-end trade execution and order lifecycle systems

Used Skills: Swift, Objective-C, WebSockets, Python (Django/Flask), microservices, Xcode Instruments, Android Profiler, APNs (Apple Push Notification service)

Find more apps in my [website](#)  .